EC38002 Econometrics I, Spring 2024

Problem Set 6: Marek Chadim[1]

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1. Given moment equations $g_i(\beta) = Z_i(Y_i - X_i'\beta)$, the criterion can be written as

$$J(\beta) = (Z'Y - Z'X\beta)'W(Z'Y - Z'X\beta).$$

The constrained problem is equivalent to finding the saddle point of the Lagrangian $L = \frac{1}{2}J(\beta) + \lambda'(R'\beta - c)$. The first order conditions are

$$\frac{\partial}{\partial\beta}L(\tilde{\beta}_{\text{cmm}}, \tilde{\lambda}_{\text{cmm}}) = -X'ZW(Z'Y - Z'X\tilde{\beta}_{\text{cmm}}) + R\tilde{\lambda}_{\text{cmm}} = 0 \tag{1}$$

$$\frac{\partial}{\partial\lambda}L(\tilde{\beta}_{\text{cmm}}, \tilde{\lambda}_{\text{cmm}}) = R'\tilde{\beta}_{\text{cmm}} - c = 0 \tag{2}$$

Premultiplying (1) by $R'(X'ZWZ'X)^{-1}$

$$R'(X'ZWZ'X)^{-1}X'ZWZ'Y - R'(X'ZWZ'X)^{-1}X'ZWZ'X\tilde{\beta}_{\text{cmm}} = R'(X'ZWZ'X)^{-1}R\tilde{\lambda}_{\text{cmm}}$$

Recognizing the unrestricted GMM estimator $\hat{\beta}_{\text{gmm}}$, imposing (2) and assuming full rank condition for invertibility to solve for $\tilde{\lambda}_{\text{cmm}}$

$$\tilde{\lambda}_{\text{cmm}} = \left[R'(X'ZWZ'X)^{-1}R\right]^{-1}(R'\hat{\beta}_{\text{gmm}} - c).$$

Substituting this expression into (1) and solving for $\tilde{\beta}_{\text{cmm}}$

$$\tilde{\beta}_{\text{cmm}} = \hat{\beta}_{\text{gmm}} - (X'ZWZ'X)^{-1}R\left[R'(X'ZWZ'X)^{-1}R\right]^{-1}(R'\hat{\beta}_{\text{gmm}} - c).$$

2. (a) The criterion function is

$$J(\beta) = \frac{1}{n}e'X\hat{\Omega}^{-1}X'e$$

rewrite using $\hat{V}_\beta = \left(\frac{1}{n}X'X\right)^{-1}\hat{\Omega}\left(\frac{1}{n}X'X\right)^{-1}$

$$J(\beta) = n\left((X'X)^{-1}X'e\right)'\hat{V}_\beta^{-1}\left((X'X)^{-1}X'e\right) = n(\beta - \hat{\beta})'\hat{V}_\beta^{-1}(\hat{\beta} - \beta),$$

since $(X'X)^{-1}X'e = (X'X)^{-1}X'(y - X'\beta) = \hat{\beta} - \beta$.

(b) Setting the efficient weight matrix from unconstrained estimation the constrained GMM estimator can be written as efficient minimum distance

$$\tilde{\beta} = \hat{\beta} - \hat{V}_\beta R\left(R'\hat{V}_\beta R\right)^{-1}(R'\hat{\beta} - 0)$$

Therefore

$$\begin{aligned} D &= n(\hat{V}_\beta(R'\hat{V}_\beta)^{-1}R'\beta)'\hat{V}_\beta^{-1}(\hat{V}_\beta(R'\hat{V}_\beta)^{-1}R'\beta) = n(\hat{V}_\beta(R'\hat{V}_\beta)^{-1}R'\beta)'R(R'\hat{V}_\beta)^{-1}R'\beta \\ &= n\beta'R(R'\hat{V}_\beta)^{-1}R'\hat{V}_\beta(R'\hat{V}_\beta)^{-1}R'\beta = n(R'\beta)(R'\hat{V}_\beta)^{-1}(R'\beta) = W_n \end{aligned}$$

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[1]

3. (a)
```
use AJR2001.dta
gen logmort2 = logmort0*logmort0
ivregress gmm loggdp (risk=logmort0 logmort2)
```

loggdp	Coefficient	std. err.
risk	.7278459	.0900634
_cons	3.336234	.6138203

(b)
```
estat overid
```
Hansen's J chi2(1) = 3.74788 (p = 0.0529)

(c)
```
ivregress 2sls loggdp (risk=logmort0 logmort2)
```

loggdp	Coefficient	Std. err.
risk	.7722554	.1130303
_cons	3.018849	.7434204

The GMM and 2SLS estimates are similar.

4. $Y_0 = 0 \implies \text{Var}[Y_0] = 0$

$$\begin{aligned}
&\text{Var}[Y_1] = \text{Var}[Y_0 + e_1] = \text{Var}[e_1] = 1,\\
&\text{Var}[Y_2] = \text{Var}[Y_1 + e_2] = \text{Var}[Y_0 + e_1 + e_2] = \text{Var}[e_1] + \text{Var}[e_2] = 2,\\
&\quad\vdots\\
&\text{Var}[Y_t] = \sum_{j=1}^{t} \text{Var}[e_j] = t.
\end{aligned}$$

The variance is not constant over time. Consequently the series is non-stationary.

5. (a) Set $e_t = 0$ for $t \neq 0$ and set $e_0 = 1$ to recursively calculate $b_j = \frac{\partial}{\partial e_t} Y_{t+j} = \frac{\partial}{\partial e_0} Y_j$.

$$\begin{aligned}
b_0 &= 1\\
b_1 &= \alpha_1 b_0 = \alpha_1\\
b_2 &= \alpha_1 b_1 = \alpha_1^2\\
&\vdots\\
b_j &= \alpha_1 b_{j-1} = \alpha_1^j.
\end{aligned}$$

(b) $\hat{b}_j(\hat{\alpha}_1) = \hat{\alpha}_1^j$ using the least squares estimator

$$\hat{\alpha} = \left(\sum_{t=2}^{n} X_t X_t'\right)^{-1} \left(\sum_{t=2}^{n} X_t Y_t\right),$$

where $X_t = (1, Y_{t-1}, \ldots, Y_{t-2})'$.

(c) By the delta method as $n \to \infty$

$$\sqrt{n}(\hat{b}_j - b_j) \xrightarrow{d} \mathcal{N}\left(0, j^2 (\hat{\alpha}_1)^{2(j-1)} s^2(\hat{\alpha}_1)\right)$$

since $\frac{\partial}{\partial \alpha_1} b_j = j\alpha_1^{j-1}$. The 95% asymptotic confidence interval is

$$\hat{b}_j \pm 1.96 \times j(\hat{\alpha}_1)^{(j-1)} s(\hat{\alpha}_1)$$

6. (a)
```
use FRED-QD.dta
gen y = 100*(pnfix/L.pnfix-1)
```

(b) Under the assumption of correct specification covariance matrix estimation is identical to the cross- section case using conventional regression methods.

```
reg y L(1/4).y, r
Linear regression                 Number of obs = 231
F(4, 226) = 25.32                   Prob > F = 0.0000
```

y	Coefficient	std. err.	t	P>|t|
L1.	.501609	.0760312	6.60	0.000
L2.	.168321	.0702794	2.40	0.017
L3.	-.0261832	.06278	-0.42	0.677
L4.	-.068269	.0525659	-1.30	0.195
_cons	.4911459	.1477817	3.32	0.001

(c) Heteroskedasticity and Autocorrelation Consistent/Robust (HAC/HAR) covariance matrix estimators are appropriate under general dependence.

```
newey y L(1/4).y, lag(5)
Regression with Newey-West standard errors
Maximum lag = 5                   Number of obs = 231
F(4,226) = 32.25                    Prob > F = 0.0000
```

y	Coefficient	std. err.	t	P>|t|
L1.	.501609	.0843377	5.95	0.000
L2.	.168321	.0692569	2.43	0.016
L3.	-.0261832	.0654151	-0.40	0.689
L4.	-.068269	.051575	-1.32	0.187
_cons	.4911459	.1425101	3.45	0.001

(d) Since the standard errors are similar the model is likely to be correctly specified in the sense that it has unforecastable errors and serially uncorrelated regression scores.

(e)
```
matrix alpha = e(b)
gen IRF = 0
replace IRF = 1 in 4
forval j = 5/14 {
    replace IRF = alpha[1, 1] * l.IRF + alpha[1, 2] * l2.IRF
    + alpha[1, 3] * l3.IRF + alpha[1, 4] * l4.IRF in `j'
}
list IRF in 5/14
```

5.	.501609	10.	.0164396
6.	.4199326	11.	-.0028752
7.	.2688901	12.	-.0087824
8.	.1241585	13.	-.0095729
9.	.0622994	14.	-.0073271